

ENTHUSIAST ADVANCE COURSE

PHASE : MED-I (B)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 14-08-2023

ANSWER KEY

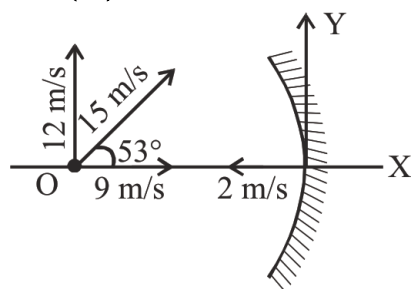
Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	2	1	2	2	2	1	4	3	2	1	3	3	2	4	3	3	4	2	1	3	4	3	3	2	2	4	3	3	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	4	4	3	3	4	2	1	3	4	4	2	1	4	2	1	1	3	3	4	4	2	1	4	4	1	2	1	4	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	2	4	2	3	1	2	3	2	3	4	4	4	3	1	1	2	3	2	1	4	3	2	1	3	3	4	1	2	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	4	3	4	1	2	4	3	1	3	4	3	1	2	2	3	1	4	2	1	1	2	4	1	3	1	1	2	1	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	4	4	1	3	1	2	2	4	4	1	2	3	3	1	1	3	2	4	2	2	4	3	2	4	2	3	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	2	2	3	4	2	4	4	2	1	2	2	4	1	3	2	1	1	4	1	2	2	2	4	2	2	1	1	4	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	4	3	1	2	2	3	2	2	4	2	1	1	4	1	1	4	4	3	4	3										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

2. Ans (2)



$$\vec{V}_o = \text{Velocity of object} = (9\hat{i} + 2\hat{j})\text{m/s}$$

$$\vec{V}_m = \text{Velocity of mirror} = -2\hat{i}\text{m/s}$$

$$m = \frac{f}{f-u} = \frac{-20}{-20 - (-30)} = -2$$

For velocity component parallel to optical axis

$$(\vec{V}_{I/m})_{\parallel} = -m^2 (\vec{V}_{o/m})_{\parallel}$$

$$(\vec{V}_{I/m})_{\parallel} = (-2)^2 11\hat{i} = -44\hat{i} \text{ m/s}$$

For velocity component perpendicular to optical axis

$$(\vec{V}_{I/m})_{\perp} = (\vec{V}_{o/m})_{\perp} = (-2)12\hat{j} = -24\hat{j}\text{m/s}$$

$$\therefore \vec{V}_{I/m} = \text{Velocity of image w.r.t. mirror}$$

$$= (\vec{V}_{I/m})_{\parallel} + (\vec{V}_{I/m})_{\perp} = (-44\hat{i} - 24\hat{j})\text{m/s}$$

$$\text{Also, } \vec{V}_{I/m} = \vec{V}_I - \vec{V}_m$$

$$\text{Or } \vec{V}_I = (-44\hat{i} - 24\hat{j}) - 2\hat{i} = (-46\hat{i} - 24\hat{j})\text{m/s}$$

6. Ans (2)

$$\frac{1}{f_{\text{net}}} = \frac{1.5 - 1}{\infty} + \frac{1 - 1.5}{10} + \frac{1.5 - 1}{-10} + \frac{1 - 1.5}{\infty}$$

$$\frac{1}{f_{\text{net}}} = \frac{-0.5}{10} - \frac{0.5}{10}$$

$$\Rightarrow \frac{1}{f_{\text{net}}} = -\frac{1}{10}$$

$$f_{\text{net}} = -10\text{cm}$$

7. Ans (1)

$$u = -25\text{cm} ; v = -75\text{cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{-75} + \frac{1}{25} = \frac{-1+3}{75} = \frac{2}{75}$$

$$P = \frac{100}{f} = \frac{200}{75} = \frac{8}{3}D$$

8. Ans (4)

$$L = v_0 + u_e = \frac{u_0 f_0}{(u_0 - f_0)} + \frac{f_e D}{f_e + D}$$

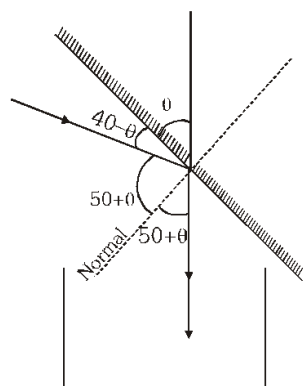
$$\Rightarrow L = \frac{2 \times 1.5}{(2 - 1.5)} + \frac{6.25 \times 25}{(6.25 + 25)} = 11\text{cm}$$

9. Ans (3)

$$\text{M.P.} = \frac{f_0}{f_e} = \frac{\beta}{\alpha} \quad [\text{Final image at } \infty]$$

$$\beta = \alpha \times \frac{f_0}{f_e} = 2^\circ \times \frac{60}{5} = 24^\circ$$

10. Ans (2)



$$\theta + 40^\circ - \theta + 50^\circ + \theta + 50^\circ + \theta = 180^\circ$$

$$\Rightarrow \theta = 20^\circ$$

Mirror is at an angle of 70° from horizontal.

12. Ans (3)

$$P_1 V_1 = P_2 V_2$$

$$(760)(0.1) = P_2(0.1 + 0.09)$$

$$P_2 = 400 \text{ mm of Hg} = 0.4 \text{ m of Hg.}$$

Hence change in pressure

$$\Delta P = P_1 - P_2 = 760 - 400 = 360 \text{ mm of Hg}$$

$$= 0.36 \text{ metre of Hg}$$

14. Ans (2)

Process ab $\rightarrow$ IT; $T = \text{constant} = U = \text{const}$;

$\Delta U = 0$ and ρ increasing

Process bc $\Rightarrow \rho = \text{constant}$ then isochoric process

Process cd $\Rightarrow$ IT process $\Rightarrow T = \text{constant}$;

$U = \text{const}$; $\Delta U = 0$ ρ decreasing

Process da $\Rightarrow \rho = \text{constant}$ then isochoric process.

18. Ans (4)

TG: @Chalnaayaaar

$$W = - \frac{R\Delta T}{\gamma - 1}$$

$$6R = - \frac{R\Delta T}{5/3 - 1}$$

$$T_f = (T - 4)K$$

19. Ans (2)

For a polytropic process PV^n

$$= \text{constant} \Rightarrow 16P_0 V_0^n = P_0 (2V_0)^n$$

$$\Rightarrow n = 4$$

$$\text{Work done} = W = \frac{P_1 V_1 - P_2 V_2}{n - 1}$$

$$= \frac{16P_0 V_0 - P_0 (2V_0)}{4 - 1} = \frac{14}{3} P_0 V_0$$

20. Ans (1)

For complete cycle

$$(\Delta U)_{12} + (\Delta U)_{23} + (\Delta U)_{31} = 0$$

$$0 + (-40) + (\Delta U)_{31} = 0$$

$$(\Delta U)_{31} = 40$$

$$W_{31} = -(\Delta U)_{31} = -40 \text{ J}$$

21. Ans (3)

$$\mu = \mu_1 + \mu_2$$

$$\frac{P(2V)}{R T_1} = \frac{P'V}{R T_1} + \frac{P'V}{R T_2} \Rightarrow \frac{2P}{R T_1} = \frac{P'}{R} \left[\frac{T_2 + T_1}{T_1 T_2} \right]$$

$$P' = \frac{2P T_2}{(T_1 + T_2)} = \frac{2 \times 1 \times 600}{(300 + 600)} = \frac{4}{3} \text{ atm}$$

22. Ans (4)

$$P_0 V = \frac{32}{32} R(400) \dots\dots\dots(1)$$

$$\frac{P_0}{2} V = \frac{m}{32} R(300) \dots\dots\dots(2)$$

Dividing (1) by (2)

$$2 = \frac{32}{m} \cdot \frac{4}{3} \Rightarrow m = \frac{64}{3} \text{ g}$$

$$\text{mass leaked} = 32 - \frac{64}{3}$$

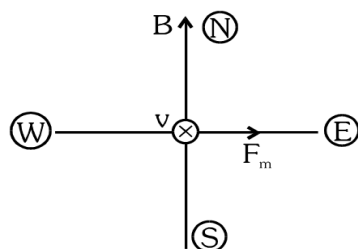
$$= \frac{32}{3} \text{ g} = 10.7 \text{ g}$$

23. Ans (3)

Ampere's law is valid for infinite wire of symmetrical current distribution.

24. Ans (3)

$\vec{v} \perp \vec{B}$, so apply right hand palm rule to find out direction of magnetic force. Hence cosmic particles deflected towards east.



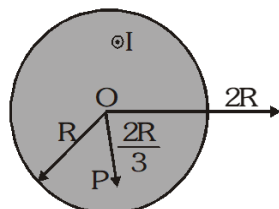
OR

$$\vec{F}_m \rightarrow (\vec{v} \times \vec{B})$$

$$\vec{F}_m \rightarrow (-\hat{k} \times \hat{j}) = \hat{i} \text{ east}$$

25. Ans (2)

Field outside the conductor



$$5 \times 10^{-6} = \frac{\mu_0 I}{2\pi r}, \text{ where } r = 2R$$

$$5 \times 10^{-6} = \frac{\mu_0 I}{2\pi(2R)} \text{ ----- (1)}$$

Field inside the conductor :-

$$B_p = \frac{\mu_0 I r}{2\pi R^2}, \text{ where } r = \left(R - \frac{R}{3}\right) = \frac{2R}{3}$$

$$= \frac{\mu_0 I (2R/3)}{2\pi R^2} = \frac{1}{3} \left(\frac{\mu_0 I}{\pi R}\right) = \frac{1}{3} (20) \times 10^{-6}$$

$$= \frac{20}{3} \mu\text{T}$$

26. Ans (2)

The magnetic field inside the solenoid is given

$$\text{by } B = \mu_0 n i$$

Where, n = number of turns per unit length,

i = current in coil

Now, we have

$$n = \frac{N}{L} = \frac{400}{0.4} = 1000$$

$$\text{and } \mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

$$\therefore B = 4 \times 3.14 \times 10^{-7} \times 1000 \times 5$$

$$= 62800 \times 10^{-7} = 6.28 \times 10^{-3} \text{ T}$$

27. Ans (4)

$$R_p : R_o : R_a$$

$$\frac{P}{q_p B} : \frac{P}{q_o B} : \frac{P}{q_a B}$$

$$\frac{1}{q_p} : \frac{1}{q_o} : \frac{1}{q_a}$$

$$\frac{1}{q_p} : \frac{1}{q_p} : \frac{1}{2q_p}$$

$$2 : 2 : 1$$

28. Ans (3)

$$\text{Radius } r = \frac{mv}{QB} = \frac{p}{QB}$$

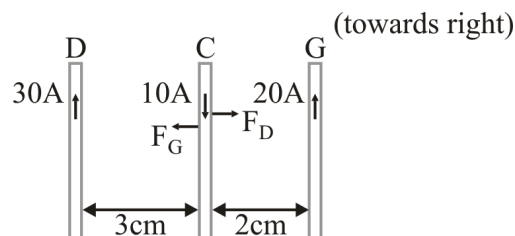
$$\text{Kinetic energy} = \frac{p^2}{2m} = \frac{(QB r)^2}{2m} = \frac{Q^2 B^2 r^2}{2m}$$

29. Ans (3)

Force on wire C due to wire D

$$F_D = 10^{-7} \times \frac{2 \times 20 \times 10}{2 \times 10^{-2}} \times 25 \times 10^{-2}$$

$$= 5 \times 10^{-4} \text{ N}$$



Force on wire C due to wire G

$$F_G = 10^{-7} \times \frac{2 \times 20 \times 10}{2 \times 10^{-2}} \times 25 \times 10^{-2}$$

$$= 5 \times 10^{-4} \text{ N} \quad (\text{towards left})$$

$$\Rightarrow \text{Net force on wire C is } F_{\text{net}} = F_D - F_G = 0$$

30. Ans (1)

$$M_1 = \left(\frac{\sqrt{3}}{4} a^2\right) I \times 4 = \sqrt{3} I a^2 \text{ (no. of turns = 4)}$$

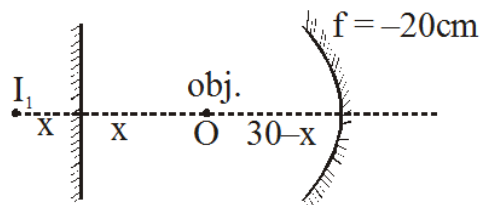
$$M_2 = a^2 I \times 3 = 3 I a^2 \text{ (no. of turns = 3)}$$

31. Ans (2)

$$M = \frac{qI\omega}{2m} \left(\frac{P_{\text{ut}}}{I} = mR^2\right)$$

SECTION-B

36. Ans (4)



1st image (I_1) is formed x behind plane mirror, it will be object for curved mirror and its image is formed on object (O) itself.

For curved mirror

$$u = -(30 + x), v = -(30 - x)$$

$$\frac{1}{-20} = \frac{-1}{(30 - x)} + \frac{1}{-(30 + x)}$$

$$\frac{1}{20} = \frac{30 + x + 30 - x}{30^2 - x^2}$$

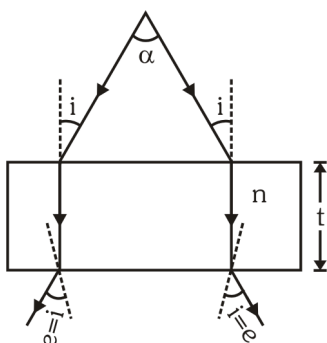
$$900 - x^2 = 60 \times 20$$

$$x^2 = -300$$

x^2 cannot be negative

37. Ans (2)

When a ray of light passes through glass slab, the emergent ray is parallel to incident ray as shown below



Also, $i = e$

So, divergent angle of the emergent beam is also same i.e. α

39. Ans (3)

Retrace path $\Rightarrow e = r_2 = 0$

$$\Rightarrow r_1 = A - r_2 = A - 0 = A = 30^\circ$$

$$1 \sin 45^\circ = \mu \sin 30^\circ \Rightarrow \mu = \sqrt{2}$$

41. Ans (4)

$$V_{\text{rms}} = \sqrt{\frac{3RT}{M}}, \text{ independent of pressure}$$

42. Ans (2)

$$|\text{Slope}| = \frac{\gamma P}{V}$$

$$\gamma_{\text{O}_2} < \gamma_{\text{He}}$$

43. Ans (1)

$$\eta_1 = 1 - \frac{T_2}{T_1} = 1 - \frac{273}{473} = \frac{200}{473}$$

$$\eta_2 = 1 - \frac{T_2}{T_1} = 1 - \frac{73}{273} = \frac{200}{273}$$

$$\text{then } \frac{\eta_1}{\eta_2} = \frac{273}{473} = 0.577$$

44. Ans (4)

$$\text{in series } K_{\text{eq}} = \frac{L_1 + L_2}{\frac{L_1}{K_1} + \frac{L_2}{K_2}} = \frac{x + 4x}{\frac{x}{K} + \frac{4x}{2K}} = \frac{5K}{3}$$

$$\frac{dQ}{dt} = \frac{\Delta T}{R_{\text{eq}}} = \frac{(T_2 - T_1)}{\frac{5x}{K_{\text{eq}}A}} = \frac{5K}{3} \frac{(T_2 - T_1)A}{5x}$$

$$\text{on comparing } f = \frac{1}{3}$$

45. Ans (2)

$$L_1 = L_2, (\theta_1 - \theta_2) \text{ is same}$$

$$\frac{K_1 A_1}{L_1} (\theta_1 - \theta_2) = \frac{K_2 A_2}{L_2} (\theta_1 - \theta_2)$$

$$K_1 A_1 = K_2 A_2$$

46. Ans (1)

$$\vec{F}_m = q\vec{v} \times \vec{B}$$

$$= (-2 \times 10^{-6}) [(2\hat{i} + 3\hat{j}) \times 10^6] \times (2\hat{j})$$

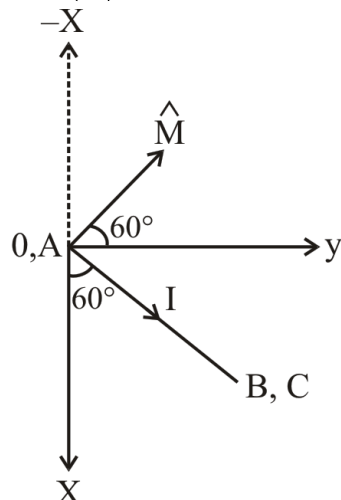
$$= -8\hat{k}\text{N}$$

i.e. 8 N along $-z$ direction

47. Ans (1)

$$\sin \theta = \frac{d}{R} = \frac{qBd}{mv}$$

48. Ans (3)



Magnetic moment of the loop can be written as,

$$\vec{M} = |\vec{M}| \hat{M} = i l^2 (\hat{M})$$

$\hat{M}$ is shown in diagram

$$\hat{M} = \frac{-\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j}$$

$$\vec{M} = \frac{i l^2}{2} (-\sqrt{3} \hat{i} + \hat{j})$$

SUBJECT : CHEMISTRY

SECTION-A

53. Ans (1)

Phosphorous show two variable covalency.

(P) in G.S. = 3

in 1st E.S. = 5

Sulphur show variable covalency.

(S) in G.S. = 2

1st E.S. = 4

2nd E.S. = 6

56. Ans (1)

NCERT Pg#105

57. Ans (2)

NCERT Pg#108

58. Ans (1)

NCERT Pg#112

59. Ans (4)

NCERT Pg#114-115

60. Ans (4)

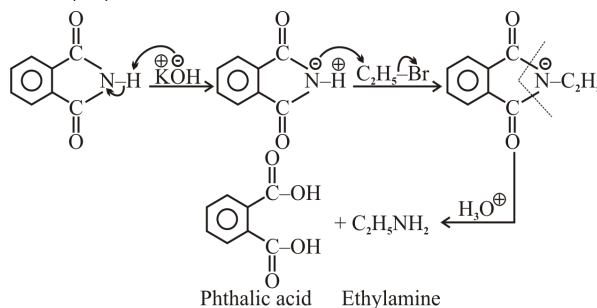
NCERT Pg#129

61. Ans (3)

NCERT Pg#129

TG: @Chalnaayaaar

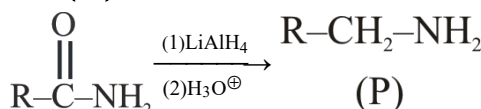
62. Ans (2)



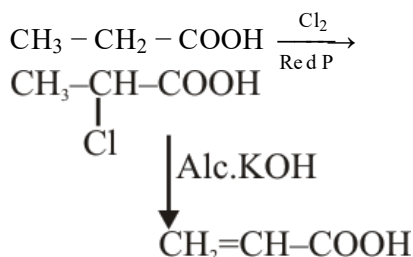
63. Ans (4)

Stephen's reaction

64. Ans (2)



65. Ans (3)



66. Ans (1)

In victor meyer test, red colouration is given by primary alcohol.

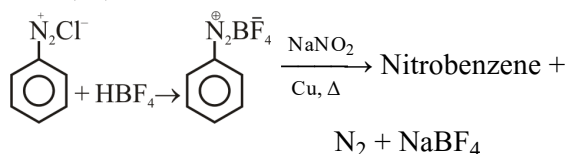
67. Ans (2)

NCERT Pg. # 393 Amine

68. Ans (3)

Phenol is less acidic than $\text{H}_2\text{CO}_3 \therefore$ do not liberate CO_2 gas on reaction with NaHCO_3

69. Ans (2)



70. Ans (3)

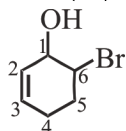
FG = different

$\therefore$ FI

71. Ans (4)

Triple bond gets lower number

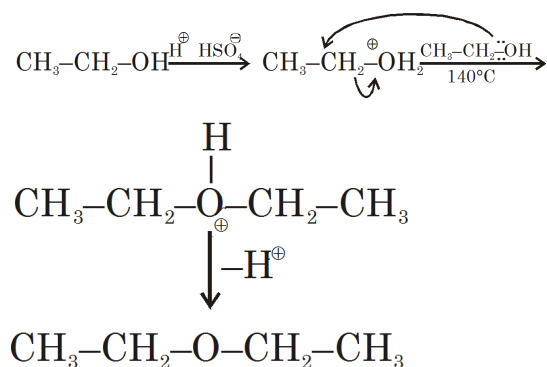
72. Ans (4)



73. Ans (4)

NCERT (XIIth) Part II, Pg. # 362

74. Ans (3)



75. Ans (1)

NCERT-XII, Part-II ; Page-375

77. Ans (2)

$$\begin{array}{ccc}
 2\text{CO}_2 & \rightleftharpoons & 2\text{CO} + \text{O}_2 \\
 2 & & 0 \quad 0 \\
 2-2x & & 2x \quad x \\
 2 - (2 \times 0.6) & & 2 \times 0.6 \quad 0.6 \\
 0.8 & & 1.2 \quad 0.6 \\
 \alpha = \frac{x}{a} = \frac{2x}{2} \Rightarrow \alpha = x = \frac{60}{100} \\
 x = 0.6
 \end{array}$$

Moles at equilibrium = 0.8 + 1.2 + 0.6 = 2.6

79. Ans (2)

Le-chatelier principle

81. Ans (4)

NCERT Pg. 210

82. Ans (3)

NCERT Pg. 208

83. Ans (2)

NCERT Pg. 205

84. Ans (1)

NCERT Pg 204

85. Ans (3)

NCERT Pg. 200

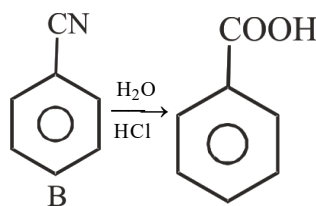
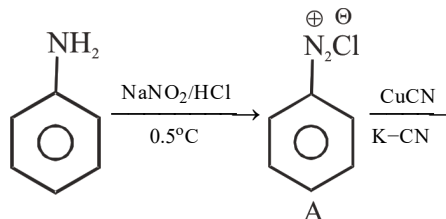
TG: @Chalgaayaaar

SECTION-B

89. Ans (2)

NCERT Pg#105

91. Ans (2)



93. Ans (3)

NCERT Reference: Class XII, Part II Pg. No. 405

94. Ans (4)

NCERT XII, Pg # 394

96. Ans (2)

NCERT Page No. # 369, (Aldehyde, Ketones & Carboxylic acid)

100. Ans (3)

NCERT Pg. 208

SUBJECT : BIOLOGY-I

SECTION-A

101. Ans (4)

NCERT XI, Pg. # 20

108. Ans (4)

NCERT-XI, Pg. # 19

109. Ans (2)

NCERT Pg. # 24

112. Ans (2)

NCERT Pg. # 24

114. Ans (1)

NCERT-XII, Pg. # 208

115. Ans (3)

NCERT-XII, Pg. # 19

116. Ans (1)

NCERT Page-19

117. **Ans (1)**

NCERT-Page No. 20

 118. **Ans (2)**

NCERT-Page No. 20

 120. **Ans (3)**

NCERT-Page No. 24

 121. **Ans (3)**

NCERT-Page No. 22-24

 122. **Ans (2)**

NCERT-Page No.21

 123. **Ans (4)**

NCERT-Page No. 45

 124. **Ans (4)**

NCERT-Page No. 23

 125. **Ans (1)**

NCERT-Page No. 24

 126. **Ans (3)**

NCERT XII, Pg. # 282

 130. **Ans (4)**

NCERT XIIth Pg# 272

 131. **Ans (4)**

NCERT (XII) Pg. # 281

 132. **Ans (1)**

NCERT XII, Pg. # 271

 133. **Ans (2)**

NCERT (XII) Pg. # 274

 134. **Ans (3)**

NCERT (XII) Pg. # 275-276

SECTION-B

 142. **Ans (2)**

NCERT Page-22

 143. **Ans (4)**

NCERT-Page No. 20

 144. **Ans (3)**

NCERT-Page No. 23

 145. **Ans (2)**

NCERT-Page No. 23

SUBJECT : BIOLOGY-II
SECTION-A

 151. **Ans (2)**

NCERT XII Pg. # 259

 152. **Ans (2)**

NCERT (XII) Pg # 237

 154. **Ans (3)**

NCERT (XII) Pg # 234

 156. **Ans (2)**

NCERT (XII) Pg # 232

 159. **Ans (2)**

NCERT XII Pg. # 266

 164. **Ans (1)**

NCERT XII, Pg # 118; Para : 6.9

 170. **Ans (1)**

NCERT Pg. # 72

 172. **Ans (2)**

NCERT, Page#70

 182. **Ans (3)**

NCERT-XII, Pg # 70 (E) & Pg # 78 (H)

SECTION-B

 186. **Ans (3)**

NCERT (XII) Pg # 267

 187. **Ans (2)**

NCERT XII Pg. # 265

 188. **Ans (2)**

NCERT (XII) Pg # 250

 190. **Ans (2)**

NCERT (XII) Pg # 233

 191. **Ans (1)**

NCERT XII Pg. # 265

 192. **Ans (1)**

NCERT Pg#117

 193. **Ans (4)**

NCERT (XII) Pg. # 115

 196. **Ans (4)**

NCERT-XII Pg. # 76